

Homework 5: Hypothesis Testing and Monte Carlo

BEE 4850/5850, Spring 2025

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Due Date

Friday, 4/17/26, 9:00pm

Overview

Instructions

The goal of this homework assignment is to practicing approaches to extreme value analysis and using information criteria to assess evidence for models.

- Problem 1 asks you to conduct a null hypothesis test using model simulations.
- Problem 1 asks you to use p -values to compare two data-generating hypotheses.
- Problem 3 asks you to use Monte Carlo optimization to identify bidding strategies for The Showcase from The Price is Right.
- Problem 4 asks you to use Monte Carlo optimization to identify house elevation strategies to mitigate flooding damages.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/Documents/GitHub/hw05-LucaKatzen1`

The following packages are included in the environment (to help you find other similar packages in other languages). The code below loads these packages for use in the subsequent notebook (the desired functionality for each package is commented next to the package).

```
using Random # random number generation and seed-setting
using DataFrames # tabular data structure
using DataFramesMeta # API which can simplify chains of DataFrames transformations
using CSV # reads/writes .csv files
using Distributions # interface to work with probability distributions
using Plots # plotting library
using StatsBase # statistical quantities like mean, median, etc
using StatsPlots # some additional statistical plotting tools
using Dates
```

Problems

Scoring

- Problem 1 is worth 3 points;
- Problem 2 is worth 5 points;
- Problem 3 is worth 7 points;
- Problem 4 is worth 10 points.

Problem 1 (3 points)

Revisit [Problem 1 from Homework 1](#).

The underlying question we would like to address is: what is the influence of drinking beer on the likelihood of being bitten by mosquitoes? There is a mechanistic reason why this might occur: mosquitoes are attracted by changes in body temperature and released CO₂, and it might be that drinking beer induces these changes. We'll analyze this question using (synthetic) data which separates an experimental population into two groups, one which drank beer and the other which drank only water.

Now we would like to **test** the null hypothesis that drinking beer makes no difference in attracting mosquitoes. Repeat the analysis from HW1. What is the p -value of the observed data under this null hypothesis? If you are interested in testing statistical significance of the proposed alternative hypothesis that beer attracts mosquitoes at the 95% confidence level, what could you conclude that this effect? In other words:

- is the alternative hypothesis statistically significant at this confidence level?
- is the observation more consistent with the alternative hypothesis than the null hypothesis?
- what does your finding imply about any conclusions you might draw about this proposed effect?

```
#Problem 1
data = DataFrame(CSV.File(joinpath(@__DIR__, "data", "bites.csv")))

beer = data[data.group .== "beer", :bites]
water = data[data.group .== "water", :bites]

observed_difference = mean(beer) - mean(water)
@show observed_difference;

observed_difference = mean(beer) - mean(water)
totalbites = vcat(beer, water)
len_beer = length(beer)
sims = 50_000
simdif = zeros(sims)

for i in 1:sims
    shuffled = shuffle(totalbites)
    simbeer = shuffled[1:len_beer]
    simwater = shuffled[len_beer+1:end]
    simdif[i] = mean(simbeer) - mean(simwater)
end

p_value_p1 = mean(simdif .>= observed_difference)
@show p_value_p1

histogram(
    simdif,
    bins = 50,
    normalize = true,
    xlabel = "Difference (Beer-Water)",
    ylabel = "Density",
```

```

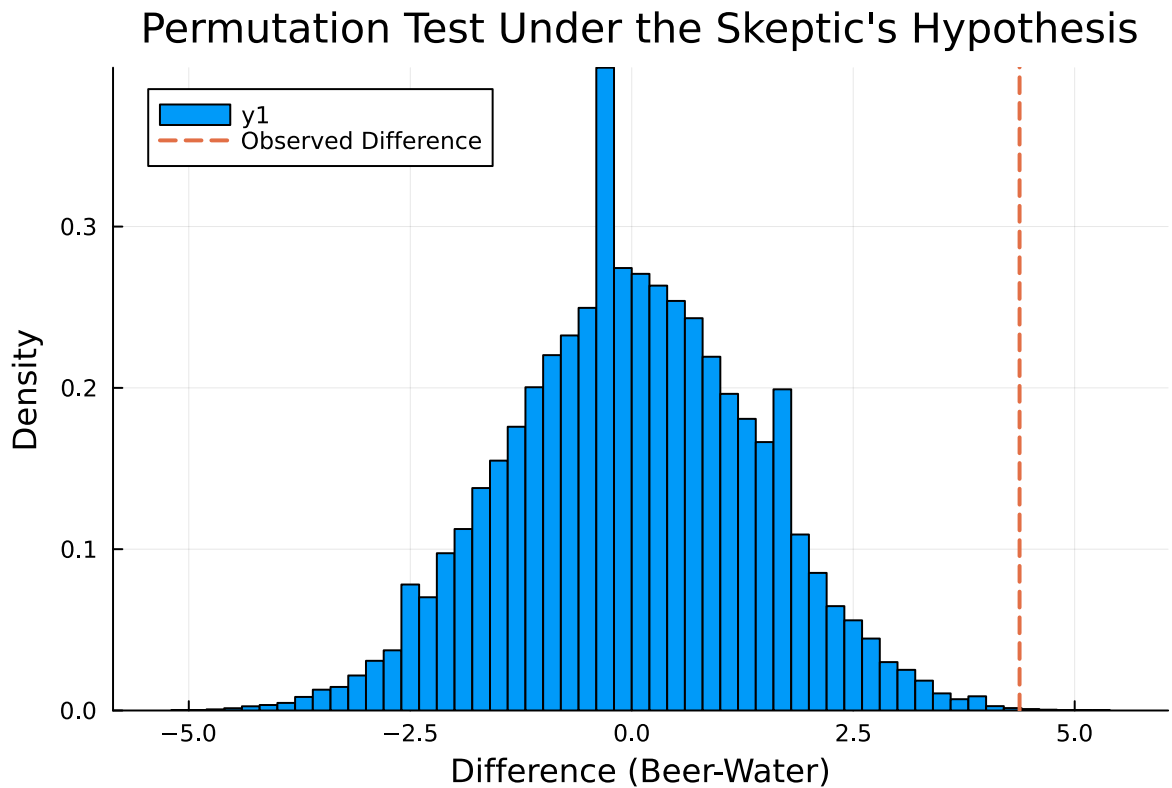
    title = "Permutation Test Under the Skeptic's Hypothesis"
)
vline!([observed_difference],
       linestyle = :dash,
       linewidth = 2,
       label = "Observed Difference")

```

```

observed_difference = 4.37777777777778
p_value_p1 = 0.00046

```



Problem 1

The p-value of the observed difference under the null hypothesis is the fraction of differences at least as large as the observed difference. Based on the results from this dataset, the p-value falls well below 0.05, showing that the observed difference is statistically significant at the 95% confidence level. This means that we can reject the null hypothesis that beer makes no difference on the likelihood of getting bit by mosquitoes. The observation is more consistent with the alternative hypothesis that beer attracts mosquitoes. However, statistical significance

alone does not establish causation, it just tells us that the observed effect is unlikely to be due to chance under the null and more testing is needed to prove the mechanistic hypothesis.

Problem 2 (5 points)

Revisit [Problem 5 from Homework 1](#).

You are trying to detect how prevalent cheating was on an exam. You are skeptical of the efficacy of just asking the students if they cheated. You are also concerned about privacy — your goal is not to punish individual students, but to see if there are systemic problems that need to be addressed. Someone proposes the following interview procedure, which the class agrees to participate in:

Each student flips a fair coin, with the results hidden from the interviewer. The student answers honestly if the coin comes up heads. Otherwise, if the coin comes up tails, the student flips the coin again, and answers “I did cheat” if heads, and “I did not cheat”, if tails.

You have a hypothesis that cheating was not prevalent, and the proportion of cheaters was no more than 5% of the class; in other words, we expect 5 “true” cheaters out of a class of 100 students. Our TA is more jaded and thinks that cheating was more rampant, and that 30% of the class cheated. The proposed interview procedure is noisy: the interviewer does not know if an admission means that the student cheated, or the result of a heads. However, it gives us a data-generating process that we can model and analyze for consistency with our hypothesis and that of the TA.

Repeat this analysis and suppose that your procedure turned up 40 “yes” answers. What is the p -value of having seen this data under both hypotheses? If you wanted to draw a conclusion at a 95% confidence level, what could you conclude about the level of cheating in the class?

```
#Problem 2
function rr_pmf(p_cheat::Float64, n::Int)
    p_yes = 0.5 * p_cheat + 0.25
    return Binomial(n, p_yes)
end

n_students = 100
observed_yes = 40
p_cheat_null = 0.05
p_cheat_alt = 0.30

dist_null = rr_pmf(p_cheat_null, n_students)
```

```

dist_alt = rr_pmf(p_cheat_alt, n_students)

p_val_null = 2 * min(cdf(dist_null, observed_yes), 1 - cdf(dist_null, observed_yes - 1))
p_val_alt = 2 * min(cdf(dist_alt, observed_yes), 1 - cdf(dist_alt, observed_yes - 1))

@show p_val_null
@show p_val_alt

ks = 0:n_students
plot(ks, pdf.(dist_null, ks),
     label = "Null (p_cheat = $(p_cheat_null))",
     xlabel = "Number of 'yes' answers",
     ylabel = "Probability",
     title = "Randomized Response: Likelihood Under Each Hypothesis",
     linewidth = 2)
plot!(ks, pdf.(dist_alt, ks),
      label = "TA hypothesis (p_cheat = $(p_cheat_alt))",
      linewidth = 2)
vline!([observed_yes],
       linestyle = :dash,
       linewidth = 2,
       label = "Observed ($(observed_yes) yes)")

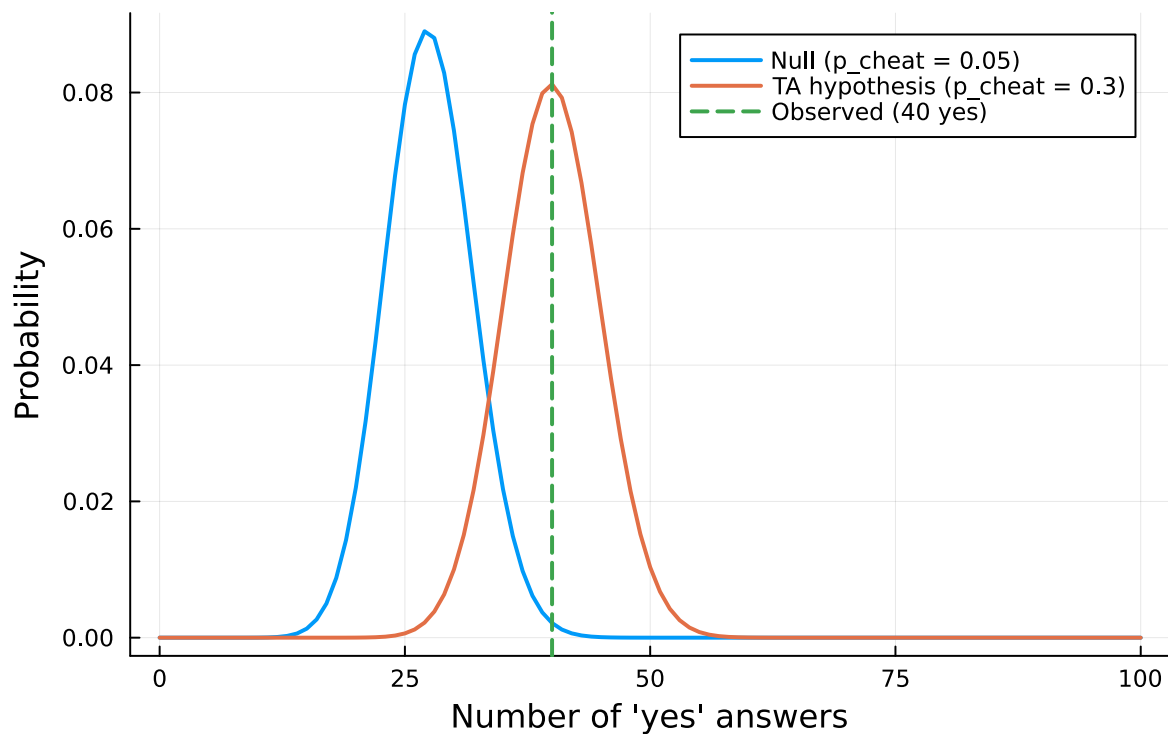
```

```

p_val_null = 0.009184831504963187
p_val_alt = 1.0758493182280748

```

Randomized Response: Likelihood Under Each Hypothesis



Problem 2.2

Under the null hypothesis, the p-value for observing 40 “yes” answers is very small (below .05), meaning the data is inconsistent with only 5% of the class having cheated. We reject the null at the 95% confidence level based on the p-value. Under the TA’s hypothesis (30% cheating), the p-value is much larger, and we fail to reject it as the data is consistent with 30% cheating. The conclusion at 95% confidence is that cheating was more likely than at 5%, and the data agrees more with the TA’s view than with the hypothesis.

Problem 3 (7 points)

[The Showcase](#) is the final round of every episode of [The Price is Right](#), matching the two big winners from the episode. Each contestant is shown a “showcase” of prizes, which are usually some combination of a trip, a motor vehicle, some furniture, and maybe some other stuff. They then each have to make a bid on the retail price of the showcase. The rules are:

- an overbid is an automatic loss;
- the contest who gets closest to the retail price wins their showcase;

- if a contestant gets within \$250 of the retail price and is closer than their opponent, they win both showcases.

You have been selected to appear on the show. As preparation, you have documented historical prizes and their values. Once you're shown the showcase, you realize that the distribution of values for this spread of prizes has historically followed a truncated normal distribution with mean \$31,000 and standard deviation \$4,500, truncated between \$15,000 and \$42,000.

You can make the following assumptions about the probability of winning and payouts:

- If you bet the exact amount, you win with probability 1 and win both showcases; the value is the double of the single showcase value.
- If you did not win both showcases but bid under the showcase value, the probability of being outbid increases linearly as the distance between your bid and the value increases (in other words, if you bid the exact value, you win with probability 1, and if you bid \$0, you win with probability 0).
- If you overbid, you automatically lose, with value \$0.

Problem 3.1

Using Monte Carlo simulation, find the wager that maximizes your probability of winning. You will need to write a function which computes the winning probability for a given bet, then optimize that function. Make sure to justify the sample size that you use in your Monte Carlo analysis.

Problem 3.2

Using Monte Carlo simulation, find the wager that maximizes your expected winnings. Is this the same as in Problem 3.1? Why do you think it is or is not? Make sure to justify the sample size that you use in your Monte Carlo analysis.

```
showcase_dist = truncated(Normal(31_000, 4_500), 15_000, 42_000)
N_showcase = 100_000
Random.seed!(42)
showcase_vals = rand(showcase_dist, N_showcase)

function showcase_pwin(bid::Float64, vals::Vector{Float64})
    wins = map(vals) do v
        bid > v ? 0.0 : 1.0 - (v - bid) / v
    end
    return mean(wins)
end
```



```

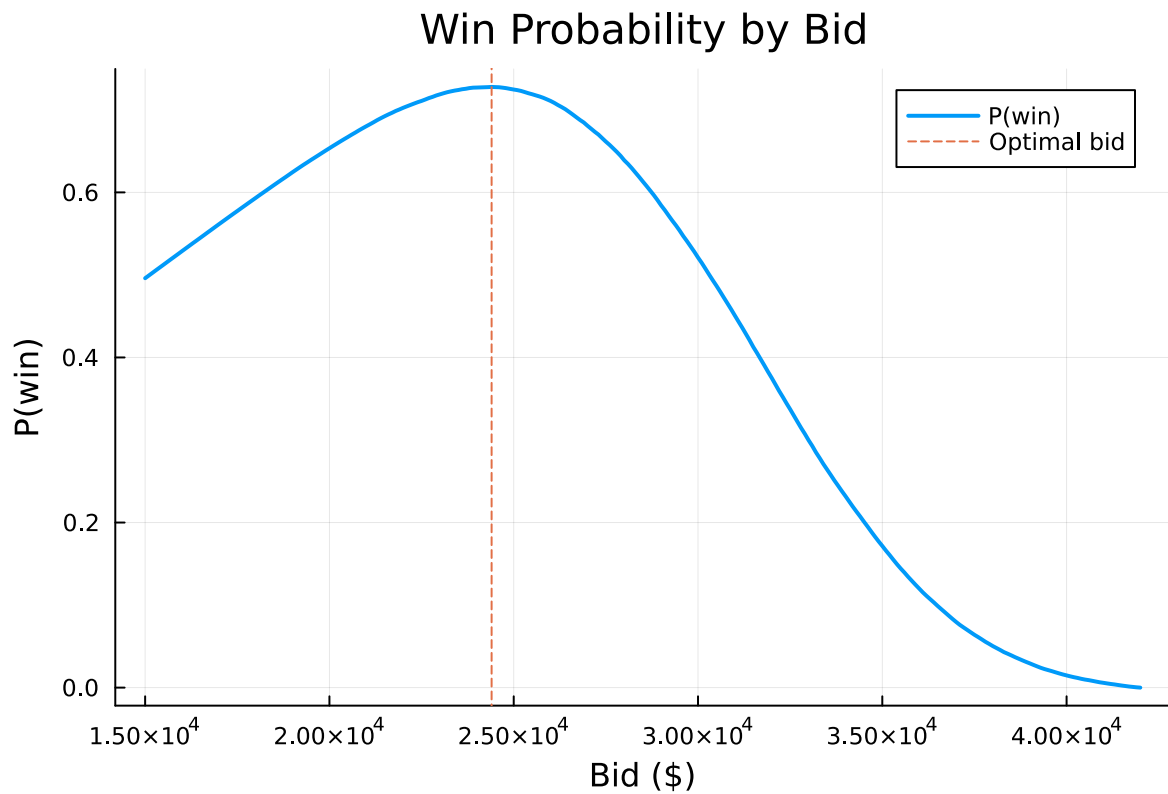
bids = collect(15_000.0:100.0:42_000.0)
pwin_vals = [showcase_pwin(b, showcase_vals) for b in bids]
best_pwin_bid = bids[argmax(pwin_vals)]

@show best_pwin_bid

plot(bids, pwin_vals,
     xlabel = "Bid (\$)",
     ylabel = "P(win)",
     title = "Win Probability by Bid",
     label = "P(win)",
     linewidth = 2)
vline!([best_pwin_bid],
       linestyle = :dash,
       label = "Optimal bid")

```

```
best_pwin_bid = 24400.0
```



Problem 3.1

The bid maximizing win probability is below the mean of the showcase distribution ($\$24400 < \31000). This is because the linear win probability function becomes maximized by getting as close to the true value as possible without overbidding. Bidding near the mean balances the risk of overbidding with the benefit of being close.

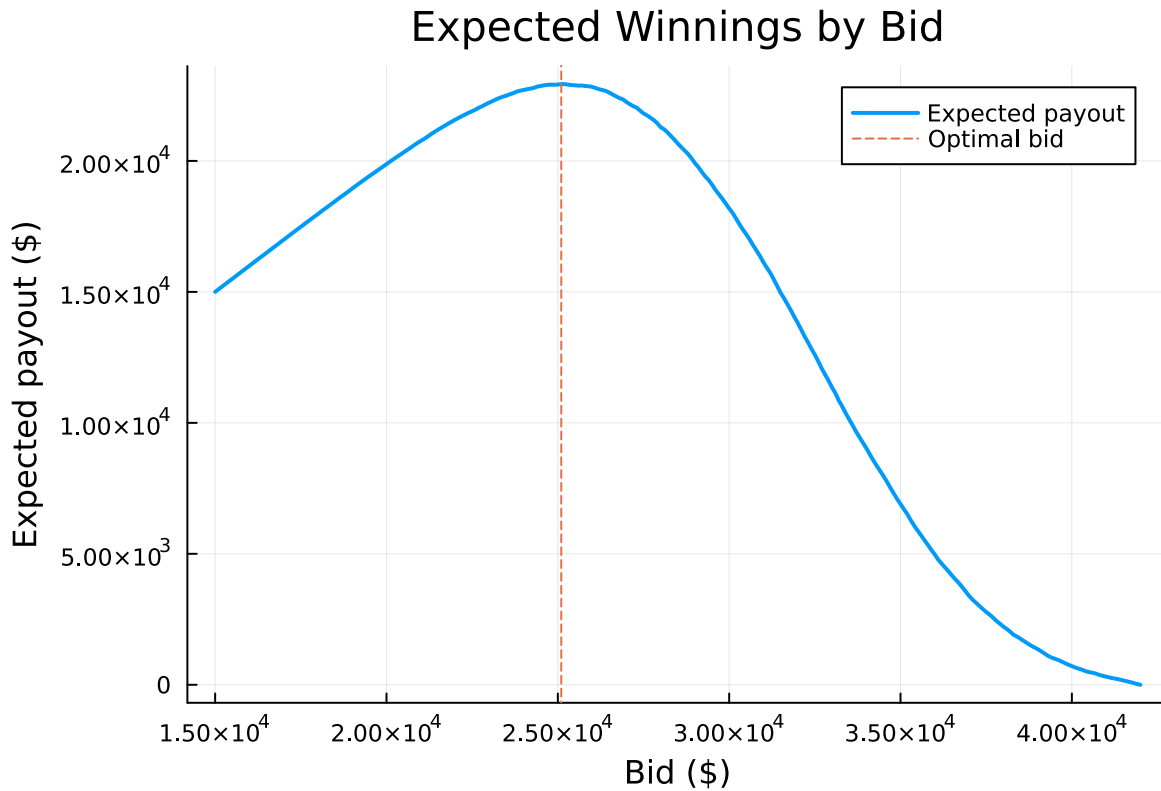
```
#Problem 3.2
function showcase_ev(bid::Float64, vals::Vector{Float64})
    payouts = map(vals) do v
        if bid > v
            0.0
        elseif bid == v
            2.0 * v
        else
            p_win = 1.0 - (v - bid) / v
            if abs(bid - v) <= 250
                2.0 * v * p_win
            else
                v * p_win
            end
        end
    end
    return mean(payouts)
end

ev_vals = [showcase_ev(b, showcase_vals) for b in bids]
best_ev_bid = bids[argmax(ev_vals)]

@show best_ev_bid

plot(bids, ev_vals,
     xlabel = "Bid (\$)",
     ylabel = "Expected payout (\$)",
     title = "Expected Winnings by Bid",
     label = "Expected payout",
     linewidth = 2)
vline!([best_ev_bid],
      linestyle = :dash,
      label = "Optimal bid")
```

```
best_ev_bid = 25100.0
```



Problem 3.2

The bid maximizing expected winnings (\$25100) is higher than the bid maximizing win probability (\$24400) because the expected value calculation weights outcomes by payout size. Bidding higher brings you closer to the true showcase value, increasing both win probability and the chance of landing within \$250 to win both showcases, doubling the payout. Even though a higher bid carries more overbid risk, the asymmetric upside of winning both showcases pulls the EV-optimal bid above the pure win-probability-maximizing bid.

Problem 4

You have been asked by a client to assess the risks of flooding to their home (which is valued at \$400,000 and only floods when the stream water levels exceed 1.5m). They would like to know whether it would be cost-effective to elevate their home to reduce flood risks. After some hydrologic analysis, you have developed a 40-year record of annual maximum water levels (in m) at the nearby stream.

You also have a depth-damage function for the fraction of the home's value which is damaged at varying flood depths:

$$d(h) = \mathbb{I}[h > 0] \frac{1}{1 + \exp(-k(x - x_0))},$$

where $k = 1.25$ and $x_0 = 2$ [1]. The graph of this function is given in the figure below.

[1] As a reminder, the indicator function $\mathbb{I}[h > 0]$ is 0 when the condition is not satisfied (in this case, when $h = 0$) and 1 when it is satisfied.

In this problem, you will need to calculate the **net present value (NPV)** of flooding damages, which converts damages over time to a present value using a discount rate[1] For example, if your discount rate is 4%, a dollar next year is worth the equivalent of about 96 cents today. More generally, with a discount rate of γ (as a decimal), a dollar of benefits in t years is worth $\$(1 - \gamma)^t$ today.

The NPV of a sequence of money x_t with discount rate γ is the sum of all of the discounted values, that is,

$$NPV = \sum_{t=0}^T x_t (1 - \gamma)^t,$$

where T is the time horizon. We'll use a discount rate of 4% in this problem, which is typical for this type of problem, and a design horizon of 30 years.

For this problem, we will assume that the cost of elevating the house Δh m is $C(\Delta h) = \mathbb{I}[h > 0](100,000 + 2,000\Delta h)$.

Problem 4.1

Fit a GEV distribution to the data in `water_depths.csv`. Plot a histogram of the data and the fitted distribution. How well does this distribution fit the data?

Problem 4.2

Use Monte Carlo simulation to estimate the NPV of the benefits of elevation levels between 0 and 5m (you don't have to consider increments finer than 0.5m). What elevation heights pass a cost-benefit test (*e.g.* the net benefits of elevation (benefits of avoiding flooding minus cost of elevation) are greater than 0)? You can assume that the costs of elevation are all up front (that is, in year 0). What height (including a possible elevation of 0m) maximizes the NPV of net benefits?

[1] Discount rates reduce future monetary values to reflect the time-value of money; that is, you would rather have a bit less money today than none today and more next year, as you could save or invest that money. The actual choice of discount rates is the subject of much economic theory and plays an important role in environmental decision-making, particularly

for multi-generational investments such as climate mitigation. For example, with a relatively large discount rate (such as 7%), all costs and benefits after a decade are effectively zero, which means it never appears cost-effective to e.g. reduce fossil fuel emissions.

```
#Problem 4.1
data2 = DataFrame(CSV.File(joinpath(@__DIR__, "data", "water_depths.csv")))

function gev_negloglik(params, data2)
    mu, log_sigma, xi = params
    sigma = exp(log_sigma)
    try
        d = GeneralizedExtremeValue(mu, sigma, xi)
        ll = sum(logpdf(d, x) for x in data)
        return isfinite(ll) ? -ll : Inf
    catch
        return Inf
    end
end

function nelder_mead(f, x0; tol=1e-8, max_iter=10_000)
    n = length(x0)
    alpha, beta, gamma, delta = 1.0, 2.0, 0.5, 0.5
    simplex = [copy(x0) for _ in 1:n+1]
    for i in 2:n+1
        simplex[i][i-1] += abs(x0[i-1]) > 0 ? 0.05 * x0[i-1] : 0.05
    end
    fvals = [f(s) for s in simplex]
    for _ in 1:max_iter
        order = sortperm(fvals)
        simplex, fvals = simplex[order], fvals[order]
        maximum(fvals) - minimum(fvals) < tol && break
        centroid = mean(simplex[1:n])
        xr = centroid + alpha * (centroid - simplex[end])
        fr = f(xr)
        if fr < fvals[1]
            xe = centroid + beta * (xr - centroid)
            fe = f(xe)
            simplex[end], fvals[end] = fe < fr ? (xe, fe) : (xr, fr)
        elseif fr < fvals[end]
            simplex[end], fvals[end] = xr, fr
        else
            xc = centroid + gamma * (simplex[end] - centroid)
            fc = f(xc)
        end
    end
end
```

```

        if fc < fvals[end]
            simplex[end], fvals[end] = xc, fc
        else
            for i in 2:n+1
                simplex[i] = simplex[1] + delta * (simplex[i] - simplex[1])
                fvals[i] = f(simplex[i])
            end
        end
    end
end
return simplex[1]
end

x0 = [mean(depth_data), log(std(depth_data)), 0.1]
result = nelder_mead(p -> gev_negloglik(p, depth_data), x0)
gev_fit = GeneralizedExtremeValue(result[1], exp(result[2]), result[3])
@show gev_fit

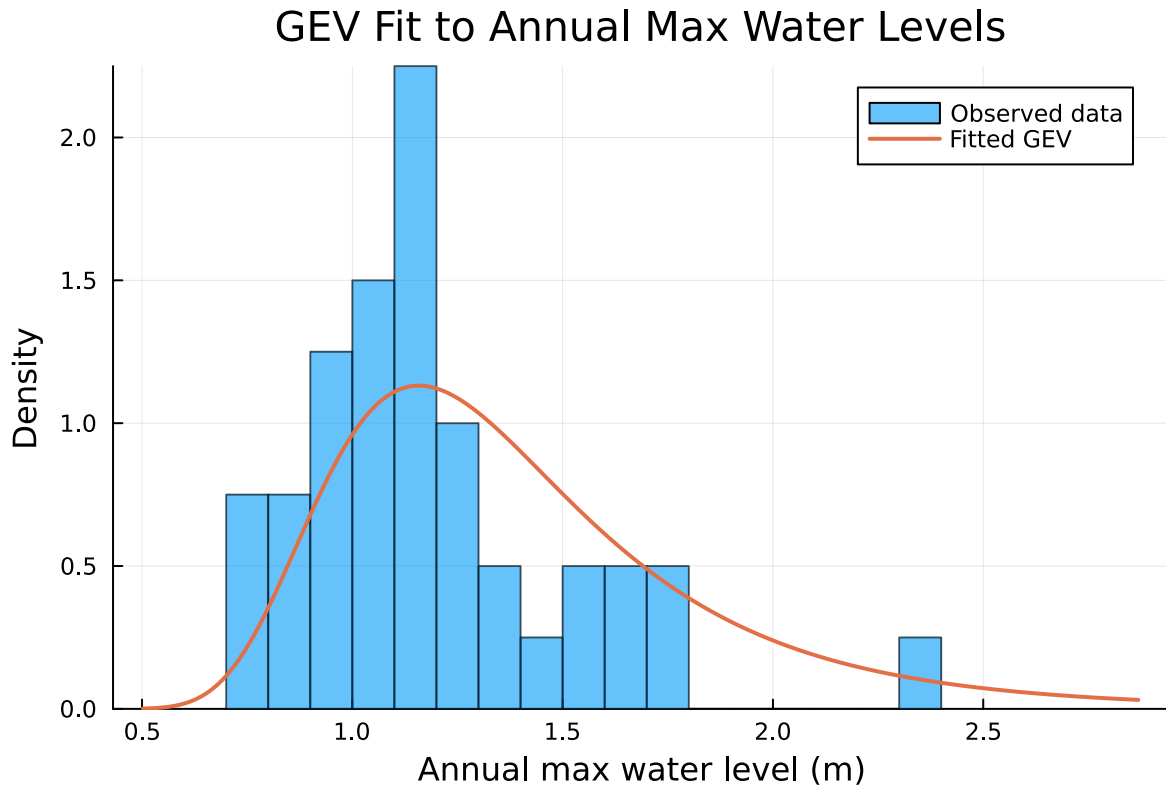
histogram(depth_data,
    bins = 20,
    normalize = true,
    xlabel = "Annual max water level (m)",
    ylabel = "Density",
    title = "GEV Fit to Annual Max Water Levels",
    label = "Observed data",
    alpha = 0.6)
xs = range(minimum(depth_data) - 0.2, maximum(depth_data) + 0.5, length=300)
plot!(xs, pdf.(gev_fit, xs),
    linewidth = 2,
    label = "Fitted GEV")

```

```

gev_fit = GeneralizedExtremeValue{Float64}(-1.1890194278903605, 0.3267202036834026, 0.1)

```



Problem 4.1

The GEV distribution fits the annual maximum water level data somewhat well because the data follows extreme value behavior. Visual inspection of the histogram overlay shows that the fitted distribution captures the tail behavior fairly well accounting for the point around 2.4 meters. Because ξ is positive (0.1), the distribution has a heavy tail. This implies that there is a non-negligible probability of very high water levels. The slight mismatch in the upper tail is important for flood risk, since those rare extreme events drive most of the damage, so the GEV distribution slightly fails on this front as it doesn't capture the magnitude of those extreme events well.

```
#Problem 4.2
```

```
home_value = 400_000.0
flood_thresh = 1.5
discount_rate = 0.04
horizon = 30
N_flood = 50_000
```

```

function depth_damage(depth_above_floor::Float64)
    depth_above_floor <= 0.0 && return 0.0
    return 0.0068 * depth_above_floor + 0.14 * depth_above_floor^2
end

elevation_cost(e::Float64) = 20_000.0 * e

function npv_damages(elevation::Float64, rng_seed::Int)
    Random.seed!(rng_seed)
    yearly_levels = rand(gev_fit, N_flood)
    n_scenarios = N_flood ÷ horizon
    total_npv = 0.0
    for s in 1:n_scenarios
        years = yearly_levels[(s-1)*horizon+1 : s*horizon]
        npv = sum(depth_damage(max(y - flood_thresh - elevation, 0.0)) * home_value /
            (1 + discount_rate)^t for (t, y) in enumerate(years))
        total_npv += npv
    end
    return total_npv / n_scenarios
end

elevations = 0.0:0.5:5.0
npv_benefits_vec = Float64[]
net_benefits_vec = Float64[]
npv_baseline = npv_damages(0.0, 1)

for e in elevations
    npv_e = npv_damages(e, 1)
    benefit = npv_baseline - npv_e
    cost = elevation_cost(e)
    net_benefit = benefit - cost
    push!(npv_benefits_vec, benefit)
    push!(net_benefits_vec, net_benefit)
end

elevations_vec = collect(elevations)
pass_cost_benefit = elevations_vec[net_benefits_vec .> 0]
best_elevation = elevations_vec[argmax(net_benefits_vec)]

@show pass_cost_benefit
@show best_elevation

```



```

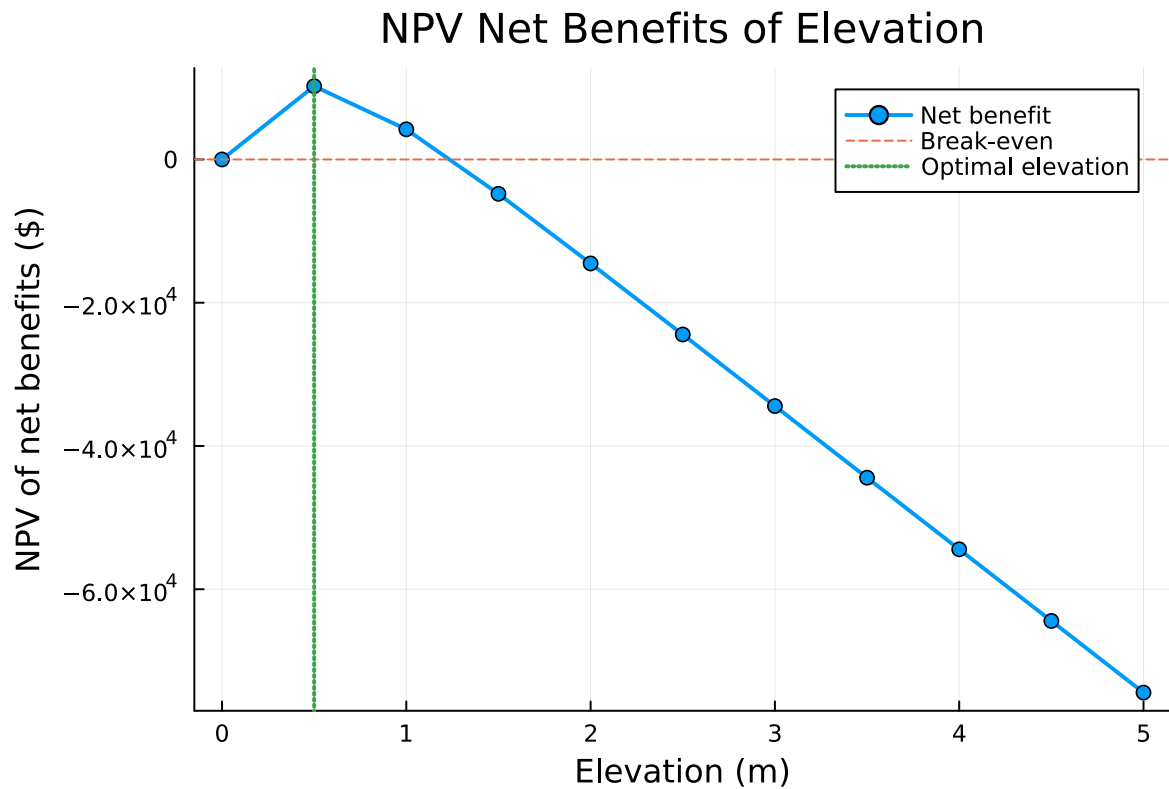
plot(elevations_vec, net_benefits_vec,
     xlabel = "Elevation (m)",
     ylabel = "NPV of net benefits (\$)",
     title = "NPV Net Benefits of Elevation",
     label = "Net benefit",
     linewidth = 2,
     marker = :circle)
hline!([0.0], linestyle = :dash, label = "Break-even", linewidth = 1)
vline!([best_elevation],
      linestyle = :dot,
      label = "Optimal elevation",
      linewidth = 2)

```

```

pass_cost_benefit = [0.5, 1.0]
best_elevation = 0.5

```



Problem 4.2

Not all elevation levels pass the cost-benefit test. At low elevations, the cost is modest but so is the flood reduction benefit (as most damaging floods already exceed low elevations). At high elevations, the cost grows linearly while the minor reduction in flood damage decreases. This is because the GEV distribution assigns low probability to extreme events that would still overtop a high elevation. The optimal elevation is the one where the marginal reduction in expected NPV of damages just equals the marginal cost, in this case .5 meters. Elevations beyond the optimal provide diminishing returns as the house is already protected against nearly all probable flood events, and the additional housing cost is not justified by further flood risk reduction.

Important to note how risk is defined here - if risk is defined as it is typically (hazard, exposure, vulnerability) this only quantifies hazard and exposure, not vulnerability which is dependent on home construction.